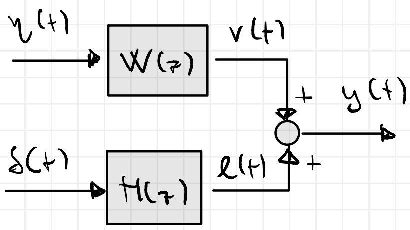
Exercise



$$W(z) = \frac{1 + 2z^{-1}}{1 - \frac{1}{4} z^{-1}}$$

$$H(z) = \frac{-2 \cdot z^{-1}}{1 - \frac{1}{4} z^{-1}}$$

WHERE $\eta(t)$ AND $\delta(t)$ ARE TWO STOCHASTIC PROCESSES SUCH THAT:

$$\begin{bmatrix} \eta(t) \\ \delta(t) \end{bmatrix} \sim WN(\mu, \sigma) \qquad \mu = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \sigma = \begin{bmatrix} 1 & \beta \\ \beta & 2 \end{bmatrix}$$

$\sim \mu_\eta$ $\qquad$ $\sim \gamma_\eta(0) = Var[\eta(t)]$

μ_δ $\qquad$ $\gamma_{\eta\delta}(0)$ $\qquad$ $\gamma_\delta(0) = Var[\delta(t)]$

$$\gamma_\eta(\tau > 0) = 0$$
$$\gamma_\delta(\tau > 0) = 0$$
$$\gamma_{\eta\delta}(\tau > 0) = 0$$

IF $\beta = 0 \Rightarrow$ THE PROCESSES ARE ORTHOGONAL $\eta(t) \perp \delta(t)$

$\Rightarrow$ PROVIDE THE PROBABILISTIC DESCRIPTION OF $y(t)$ WHEN $\beta = 0$; COMPUTE THE VARIANCE OF $y(t)$

- MEAN VALUE

$$\mathbb{E}[y(t)] = \mathbb{E}[v(t)] + \mathbb{E}[e(t)] = W(z=1) \cdot \mathbb{E}[\underbrace{\eta(t)}_{\sim \mu_\eta}]$$

$$= \frac{3}{3/4} \cdot \mu_\eta = \boxed{4}$$

- COVARIANCE FUNCTION

$$\gamma_y(\tau) = \mathbb{E}[(y(t) - \mu_y)(y(t-\tau) - \mu_y)] = \gamma_{\tilde{y}}(\tau)$$

$\tilde{y}$: DE-BIASED PROCESS

$$\boxed{\tilde{y} = \tilde{v} + \tilde{e}}$$

- IT'S GENERATED BY THE SAME T.F. OF $y(t)$
- FED BY A ZERO MEAN NOISE

$$\gamma_{\bar y}(\tau) = E\left[\left(\bar v(t)+\bar e(t)\right)\left(\bar v(t-\tau)+\bar e(t-\tau)\right)\right]$$

$$= E[\bar v(t)\cdot\bar v(t-\tau)] + E[\bar e(t)\cdot\bar e(t-\tau)] +$$

$$+ E[\bar v(t)\cdot\bar e(t-\tau)] + E[\bar e(t)\cdot\bar v(t-\tau)]$$

$$= \gamma_{\bar v}(\tau) + \gamma_{\bar e}(\tau)$$

RECALLING THAT

$$\bar v(t) = f_v\left(\eta(t), \eta(t-1), \dots\right)$$
$$\bar e(t) = f_e\left(\delta(t), \delta(t-1), \dots\right)$$

$\Rightarrow$ RECALLING THAT $\beta = 0$ $\Rightarrow$ $\boxed{\gamma_{\eta\delta}(\tau) = 0 \ \forall \tau}$

$= 0$ SINCE THEY LOOK FOR A CORRELATION BETWEEN $\eta(t)$ AND $\delta(t)$ IN DIFFERENT TIME INSTANTS. SINCE HOLDS THIS MEANS THEY ARE EQUAL TO ZERO

$\Downarrow$

$$\gamma_{\bar y}(\tau) = \gamma_{\bar v}(\tau) + \gamma_{\bar e}(\tau)$$

▣ $\gamma_{\bar v}(\tau)$

$$\eta \longrightarrow \boxed{W(z)} \xrightarrow{\ \bar v\ } \qquad W(z) = \frac{1+2z^{-1}}{1-\frac{1}{4}z^{-1}}$$

LET'S WRITE $W(z)$ IN CANONICAL FORM (NOT MANDATORY, BUT SUGGESTED)

- SAME ORDER ✓
- MONIC ✓
- COPRIME ✓
- POLES / ZEROS ARE IN THE UNITARY CIRCLE ✗

$$\bar v(t) = \frac{1+2z^{-1}}{1-\frac{1}{4}z^{-1}}\cdot\eta(t)$$

$$= \frac{1+2z^{-1}}{1-\frac{1}{4}z^{-1}}\cdot\boxed{\frac{1+\frac{1}{2}z^{-1}}{1+2z^{-1}}\cdot 2}\cdot\eta(t)$$

DC GAIN CHECK: $\frac{2+1}{1+2} = 1$ ✓

ALL - PASS FILTER

$$= \frac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{4}z^{-1}}\cdot\tilde\eta(t) \qquad \tilde\eta(t) \sim WN\left(0\cdot 2, \overset{4}{1\cdot 2^2}\right)$$

IN TIME DOMAIN WE HAVE:

$$\boxed{\bar v(t) = 1/4\, \bar v(t-1) + \tilde\eta(t) + 1/2\, \tilde\eta(t-1)} \qquad \text{ARMA}(1,1)$$

LET'S COMPUTE THE COVARIANCE FUNCTION:

$$\gamma_{\bar v}(\tau) = \mathbb{E}[\bar v(t)\cdot \bar v(t-\tau)]$$
$$= \mathbb{E}\left[\left(1/4\, \bar v(t-1) + \tilde\eta(t) + 1/2\, \tilde\eta(t-1)\right)\cdot \bar v(t-\tau)\right]$$
$$= 1/4\, \mathbb{E}[\bar v(t-1)\cdot \bar v(t-\tau)] + \mathbb{E}[\tilde\eta(t)\cdot \bar v(t-\tau)] + 1/2\, \mathbb{E}[\tilde\eta(t-1)\cdot \bar v(t-\tau)]$$

$$\gamma_{\bar v}(\tau) = 1/4 \cdot \gamma_{\bar v}(\tau-1) + \square + 1/2\, \square$$

$\square$ $\mathbb{E}[\tilde\eta(t)\cdot \bar v(t-\tau)]$

- $\tau = 0$ $\quad \mathbb{E}[\tilde\eta(t)\cdot \bar v(t)] = \mathbb{E}\left[\tilde\eta(t)\left(1/4\, \bar v(t-1) + \tilde\eta(t) + 1/2\, \tilde\eta(t-1)\right)\right]$
  $= \lambda^2_{\tilde\eta} + 0 + 0 = 4$
- $\tau \geq 1$ $\quad 0$

$\square$ $\mathbb{E}[\tilde\eta(t-1)\cdot \bar v(t-\tau)]$

- $\tau = 0$: $\mathbb{E}\left[\tilde\eta(t-1)\cdot \overbrace{\left(1/4\cdot \bar v(t-1) + \tilde\eta(t) + 1/2\, \tilde\eta(t-1)\right)}^{\bar v(t)}\right]$
  $= 1/4\, \mathbb{E}[\tilde\eta(t-1)\cdot \bar v(t-1)] + 0 + 1/2\, \lambda^2_{\tilde\eta}$
  $= 1/4 \cdot \lambda^2_{\tilde\eta} + 1/2\, \lambda^2_{\tilde\eta} = 1 + 2 = 3$
- $\tau = 1$: $\mathbb{E}\left[\tilde\eta(t-1)\cdot \overbrace{\left(1/4\cdot \bar v(t-2) + \tilde\eta(t-1) + 1/2\, \tilde\eta(t-2)\right)}^{\bar v(t-1)}\right]$
  $= \lambda^2_{\tilde\eta} = 4$

$$\gamma_{\bar v}(\tau) = 1/4 \cdot \gamma_{\bar v}(\tau-1) + \square + 1/2\, \square$$

SO WE HAVE:

$$\begin{cases} \gamma(0) = 1/4\, \gamma(1) + 4 + 3/2 \\ \gamma(1) = 1/4\, \gamma(0) + 2 \end{cases} \Rightarrow \boxed{\gamma(0) = \frac{16}{15}\cdot 6 \qquad \gamma(1) = \frac{54}{15} \qquad \gamma(\tau) = 1/4\, \gamma(\tau-1),\ \tau \geq 2}$$

$\gamma_{\bar{e}}(\tau)$

$$\bar{\xi} \longrightarrow \boxed{H(z)} \longrightarrow \bar{e} \qquad H(z) = \frac{-2z^{-1}}{1 - 1/4\, z^{-1}} \qquad \text{ARMA}(1,1)$$

LET'S WRITE THE PROCESS IN CANONICAL FORM:

- SAME ORDER ✗
- MONIC ✗
- COPRIME
- ROOTS UNITARY CIRCLE

$$\bar{e}(t) = \frac{-2z^{-1}}{1 - 1/4\, z^{-1}} \cdot \bar{\xi}(t)$$

$$= \frac{1}{1 - 1/4\, z^{-1}} \cdot (-2z^{-1}) \cdot \bar{\xi}(t)$$

$$= \frac{1}{1 - 1/4\, z^{-1}} \cdot \tilde{\xi}(t) \qquad \tilde{\xi}(t) \sim WN(0, \overset{\sim 8}{2 \cdot 2^2})$$

$\hookrightarrow \tilde{H}(z)$ IS AR(1) PROCESS

WE CAN USE THE YULE-WALKER FORMULA:

$$\boxed{\begin{aligned} \gamma_{\bar{e}}(0) &= \frac{1}{1-(1/4)^2} \cdot \lambda^2_{\tilde{\xi}} = \frac{16}{15} \cdot 8 \\ \gamma_{\bar{e}}(\tau) &\overset{*}{=} 1/4 \cdot \gamma_{\bar{e}}(\tau - 1) \end{aligned}}$$

* CORRECTED W.R.T. THE CLASS NOTES

THE VARIANCE OF $y(t)$ IS EQUAL TO:

$$\text{Var}[y(t)] = \gamma_y(0) = \gamma_{\bar{y}}(0) = \gamma_{\bar{v}}(0) + \gamma_{\bar{e}}(0) = \frac{16}{15} \cdot 6 + \frac{16}{15} \cdot 8$$

$$= \boxed{\frac{16}{15} \cdot 14}$$

⇒ HOW THE RESULT CHANGES IF $\beta = 1$?

THE HARDEST PROBLEM IS THAT:

$$\mathbb{E}[\bar{v}(t)\cdot \bar{e}(t-\tau)]$$
$$\mathbb{E}[\bar{e}(t)\cdot \bar{v}(t-\tau)]$$

ARE NO LONGER "0" SINCE THE GENERATING NOISES ARE CORRELATED

ALTERNATIVE (INSTEAD OF COMPUTING) IDEA:

"TWO CORRELATED NOISES CAN BE DESCRIBED BY A LINEAR COMBINATION OF TWO ORTHOGONAL NOISES"

$$\begin{bmatrix} \eta \\ \delta \end{bmatrix} = \Lambda \begin{bmatrix} \gamma \\ \varepsilon \end{bmatrix}$$

WHERE

$$\gamma \sim WN(\mu_\gamma, 1)$$
$$\varepsilon \sim WN(\mu_\varepsilon, 1)$$
$$\gamma_{\gamma\varepsilon}(\tau) = 0 \quad \forall \tau$$

$$\Lambda = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- MEAN VALUE

$$\mathbb{E}\begin{bmatrix} \eta \\ \delta \end{bmatrix} = \Lambda \cdot \mathbb{E}\begin{bmatrix} \gamma \\ \varepsilon \end{bmatrix} \Rightarrow \begin{bmatrix} \mu_\eta \\ \mu_\delta \end{bmatrix} = \Lambda \begin{bmatrix} \mu_\gamma \\ \mu_\varepsilon \end{bmatrix}$$

- COVARIANCE MATRIX

$$\mathbb{E}\left[\begin{bmatrix} \eta \\ \delta \end{bmatrix}\begin{bmatrix} \eta & \delta \end{bmatrix}\right] = \Lambda \cdot \mathbb{E}\left[\begin{bmatrix} \gamma \\ \varepsilon \end{bmatrix}\begin{bmatrix} \gamma & \varepsilon \end{bmatrix}\right]\Lambda^T = \Lambda \cdot \Lambda^T$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$Var[\gamma]$

$Var[\varepsilon]$

$\gamma_{\gamma\varepsilon}(0) = 0!$

$\beta = 1$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

* SINCE THE MATRIX IS SYMMETRIC, ONE EQUATION IS REDUNDANT

$$\begin{bmatrix} a^2+b^2 & ca+bd \\ ac+bd & c^2+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

4 EQ.*, 4 UNKNOWNS (NONLINEAR)

SUPPOSE $b = 0$ $\Rightarrow$ $\begin{aligned} a^2 &= 1 \\ ca &= 1 \\ c^2 + d^2 &= 2 \end{aligned}$ $\Rightarrow$ $\begin{aligned} a &= 1^{*} \\ c &= 1 \\ d &= 1 \end{aligned}$ $\longrightarrow \Lambda = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

* NOTE: ALSO $a = -1$, $c = -1$, $d = 1$ IS AN ACCEPTABLE SOLUTION

HOW DO WE USE THIS RESULT?

$$y(t) = W \cdot \eta + H \cdot \delta = [W \quad H] \begin{bmatrix} \eta \\ \delta \end{bmatrix}$$

$$= [W \quad H] \cdot \Lambda \cdot \begin{bmatrix} \gamma \\ \varepsilon \end{bmatrix}$$

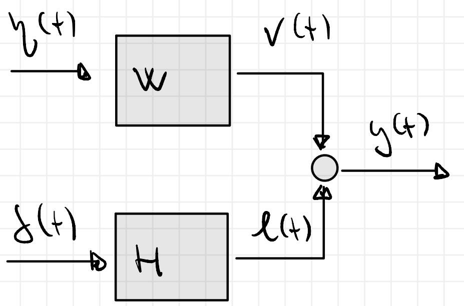
$$y(t) = W^{*} \cdot \gamma(t) + H^{*} \cdot \varepsilon(t)$$

WHERE: $W^{*} = W + H = \dfrac{1}{1 - 1/4\, z^{-1}}$ $\qquad \gamma \sim WN(\mu_\gamma, 1)$

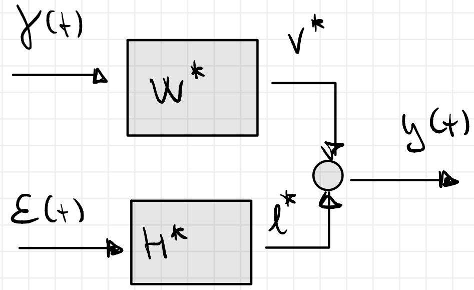
$H^{*} = H = \dfrac{-2z^{-1}}{1 - 1/4\, z^{-1}}$ $\qquad \varepsilon \sim WN(\mu_\varepsilon, 1)$

IN ORDER TO FIND THE MEAN VALUES:

$$\begin{bmatrix} \mu_\gamma \\ \mu_\varepsilon \end{bmatrix} = \Lambda^{-1} \begin{bmatrix} \mu_\eta \\ \mu_\delta \end{bmatrix} = \frac{1}{1} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$



$\Longleftrightarrow$



WITH CORRELATED NOISES

WITH UNCORRELATED NOISES

- MEAN VALUE

$$\mathbb{E}[y] = \mathbb{E}[v^{*}] + \mathbb{E}[l^{*}] = W^{*}(1) \cdot \mu_\gamma + H^{*}(1) \cdot \mu_\varepsilon$$

$$= 4/3 \cdot 1 + 2 \cdot 4/3 = \boxed{4}$$

- COVARIANCE FUNCTION

$$\gamma_y(t) = \gamma_{\bar{y}}(t) = \gamma_{\bar{v}^*}(\tau) + \gamma_{\bar{\ell}^*}(\tau) + 0$$

☒ $\gamma_{\bar{v}^*}(\tau)$ $\quad W^*(z) = \dfrac{1}{1 - 1/4\, z^{-1}}$ AR(1) $\Rightarrow$ YULE - WALKER AR(1)

$$\boxed{\begin{aligned} \gamma_{\bar{v}^*}(0) &= \frac{1}{1 - 1/16} \lambda_\gamma^2 = \frac{16}{15} \cdot 1 \\ \gamma_{\bar{v}^*}(\tau) &= 1/4\, \gamma(\tau - 1) \end{aligned}}$$

☒ $\gamma_{\bar{\ell}^*}(\tau)$ $\quad H^* = H$ THE ONLY DIFFERENCE IS THAT H IS FED WITH A NOISE WITH A DIFFERENT VARIANCE!

SO, RECALLING THE PREVIOUS CALCULATIONS WE HAVE:

$$\gamma_{\bar{e}^*}(0) = \frac{1}{1 - (1/4)^2} \cdot \lambda_{\tilde{\delta}}^2$$

$\hookrightarrow$ PREVIOUS VALUE! $4 \cdot \lambda_\delta^2$

WE NOW USE THE CORRECT NOISE VARIANCE!

$$\lambda_\delta^2 \longrightarrow \lambda_\varepsilon^2$$

$$= \frac{1}{1 - 1/16} \cdot 4 \cdot \underbrace{1}_{\lambda_\varepsilon^2}$$

$$\gamma_{\bar{e}^*}(\tau) = 1/4\, \gamma_{\bar{e}^*}(\tau - 1)$$

EVENTUALLY:

$$Var[y(t)] = \gamma_y(0) = \gamma_{\bar{y}}(0) = \gamma_{\bar{v}^*}(0) + \gamma_{\bar{e}^*}(0) = \frac{16}{15} \cdot 1 + \frac{16}{15} \cdot 4$$

$$= \boxed{\frac{16}{15} \cdot 5}$$